GPiBridge Lite: USB-C to IEEE-488 (GPIB)

XC6206: 3.3 V, 200 mA, for the RP2354A only (about 50 mA). Dropout 0.25 V at 100 mA: a USB port sagging to 4.4 V still gives 3.3 V.

Core regulator (1.1 V): Raspberry Pi's parts and layout, copied exactly (gen/rpi_minimal). L1 is polarity-marked: its dot must sit as their footprint shows.

3.3 V and 1.1 V: one 100 nF for every supply pin. On the board each sits at its pin, except where the pin is wailed in by signal lines (pins 6, 11, 20, 23): those join it through the supply rings under the chip.

BUS (green): lit while the firmware is on the bus. BUS_PWR_EN is the signal that switches GPIBridge's transceivers on, which lights the same LED there.

CC1, CC2: 5.1 k to ground say 'device': the host then gives 5 V.

USB 2.0 full speed, 12 Mbit/s. 27 R at the chip (RP2350 datasheet).

USB-C. Takes 5 V, about 40 mA: the RP2354A and its LEDs.

ABM8-272-T3, 12 MHz, 10 pF load: the crystal Raspberry Pi specifies, with their 15 pF loads and the 1 k that limits its drive.

R30-R45: 100 R in every line (R30 + the GPIO number). They hold the current to about 35 mA if an instrument drives a line high while the RP2354A holds it low, and static on a contact reaches the pin through them. A line held low sits at about 0.25 V with one instrument's terminations on it (3 k to 5 V, 6.2 k to ground), about 0.6 V with three (TTL low: under 0.8 V).

IEEE-488 plug, hand-soldered. Contacts 1-12 run along the wide side of the D. TTL levels, low = true. The RP2354A's bus pins are 5 V tolerant while it is powered (5.5 V max) and take up to 3.63 V while it is not: a standard bus idles at about 3.4 V.

Raspberry Pi debug connector (JST-SH 3-pin): the Debug Probe's 'D' cable. 100 R on the target side, at the connector (the spec: at the chip). 3.3 V logic.

BOOT: hold it while plugging in to start the RP2354A's USB bootloader. Rarely needed: 'picotool reboot -f -u' gets there from software. RUN stays open: it has its own pull-up inside the RP2354A (datasheet table 1677), and the Debug Probe resets the chip over SWD.

The 2 MB flash is inside the RP2354A: these stay open.

DC: the board ID. R21 is 1 k down here, 10 k up on GPIBridge. The firmware reads DC at start-up, before it drives anything: low runs the Lite's pin layer.

Open-drain: a line is asserted by driving its pin low and released by letting go. NRFD, NDAC and SRQ are wired-OR on every bus, and the Lite treats the other lines the same way, so it never needs to know who else is driving.

UART0 console, 115200 baud: the Debug Probe's 'U' cable. Works when USB does not.

USB (yellow) and ERR (red) are the firmware's. Both are low-voltage colours, so a 3.3 V pin lights them through 330 R (3 mA).

An RP2354A runs the IEEE-488 handshake and talks to the host over USB as a Prologix-compatible serial controller. GPiBridge Lite has no bus transceivers: each of the sixteen bus lines meets an RP2354A pin through 100 R. The firmware drives them open-drain (low to assert, let go to release), and the pads' output is forced low in hardware, so no pin can drive a line high. That carries one instrument, or a few on a short cable: 12 mA sinks against the standard's 48 mA, and no terminations of its own. For a full bus, GPiBridge. The plug hangs under the board; everything else is on top.

H1, H2: M2 holes: the two screws that close the housing pass through them.

RP2354A on the bus through 100 R: a Prologix-compatible '++' controller over USB
Hardware CERN-OHL-P-2.0, software MIT
Made by gen/schematic.py from gen/pins.py and gen/parts.py: edit those, not this file
Jason Gallicchio and Claude -- https://github.com/gallicchio/GPiBridge

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File: GPiBridgeLite.kicad_sch

Title: GPiBridge Lite: USB-C to IEEE-488 (GPIB) adapter

Size: A3	Date: 2026-10-06	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/1